SelectNorm: A Disagreement-Factorized Analysis of Per-Input Normalization for Arabizi with Frozen Encoders

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Abstract. A classifier trained on Arabic-script text can be applied to Latin-script Arabizi directly or after transliteration; the better choice varies by input. We formulate this per-input choice as a cost-aware Bayes decision that separates sunk from avoidable inference cost, and factorise its expected benefit into the probability that the two views disagree and the margin by which the normalised view then wins, so that disagreement detection and conditional advantage estimation can be measured separately. With four frozen encoders and linear heads, trained on Tunisian data, normalising improves macro-F1 by 2–19 points on Tunisian and Algerian sets, and on Moroccan text its sign depends on the encoder. Learned per-input routing performs on par with a direct-gain router (every paired 95% interval within ±0.7 points) but does not consistently exceed the better single view, and because it reads the first pass before deciding, it costs more than either single view. Disagreements are moderately predictable (AUC 0.62–0.74), whereas the winning view is weakly predictable from decision-point features (AUC at most 0.71; point estimates below 0.5 in two cases), also when a linear estimator is fitted on target labels. The normaliser's safeguards matter more than the routing components. The factorisation thus locates the bottleneck of selective normalisation in estimating the conditional advantage.

Keywords: Arabizi · Transliteration · Selective normalisation · Learning to defer · Frozen encoders · Dialect transfer

1 Introduction

North African social-media text is written in Arabic script, in Latin-script Arabizi with digits standing for Arabic consonants (*3* = ع, *7* = ح), and with French segments inside the same sentence. A classifier trained on Arabic-script data can be applied to Arabizi as it is or after transliteration, which helps when it recovers known words and harms when it garbles French segments, negation or names. The choice is usually made once per dataset [25,23,3], although the better choice varies from input to input.

We ask whether a frozen classifier can learn, without labels at inference time, *for which inputs* normalisation pays off, and whether that decision transfers from Tunisian to Algerian and Moroccan Arabizi. Post-hoc deferral rules [18] and loss predictors [20] estimate per-input benefit, but not how such a rule moves to a new dialect with scarce labels, nor which part of it fails.

With binary labels, the expected benefit of normalising equals the probability that the two views disagree times the margin by which the normalised view wins when they do (Eq. (3)). The first factor needs no labels, the second only labels on disagreements. The identity is elementary and underlies query-by-committee [30]; we use it to separate disagreement detection, conditional advantage estimation and a cost-aware decision, so that each can be measured. We ask whether per-input selection beats dataset-level choices and strong routers (RQ1) across corpora and dialects (RQ2), what it costs (RQ3), whether it agrees with predictions on human transliterations (RQ4), which factor limits it (RQ5), and how much a few target labels help (RQ6). Our contributions are:

- a formulation of the per-input choice between two views as a cost-aware Bayes decision separating sunk from avoidable cost (Sect. 3);
- a factorisation of the benefit into disagreement and conditional advantage, each measured for four encoders and three Maghrebi varieties (Sect. 6.3);
- a pre-specified comparison with confidence thresholding, direct-gain deferral, per-view loss prediction and a two-view ensemble, with corrected paired tests (Sect. 6.1);
- a cost analysis, checked by timing, of what routing can save (the second pass) and what it cannot (the first) (Sect. 6.2);
- evidence on where source estimates break down in transfer (Sect. 6.3);
- a conservative Arabizi normaliser whose safeguards matter more than the routing components (Sect. 6.4; intrinsic accuracy not independently validated).

SelectNorm performs on par with direct-gain deferral but does not consistently beat the better single view; the factorisation shows that the bottleneck is estimating the conditional advantage, not detecting disagreement.

2 Related Work

Adaptive selection and deferral. Selective classification abstains under low confidence [12,13]; learning to defer trains a classifier and a deferral rule jointly [24]. Closest to our setting are post-hoc cascade rules: Jitkrittum et al. [18] show that confidence-based deferral is optimal only under uniform second-stage errors and estimate the deferral gain directly; Gupta et al. [16] learn deferral from token-level uncertainty. LLM routers choose a model from preference data [27] or from predicted weak–strong disagreement [26]. These methods choose between *models*; SelectNorm chooses between two *views* for one frozen model and factorises the gain.

Input transformation and multilingual processing. Kim et al. [20] predict the loss of each test-time augmentation (our B4b). Kang et al. [19] learn when to translate a query before LLM reasoning. Translate-test is usually chosen per task rather than per input [3], and transliterating unseen scripts can help multilingual encoders [25,23]. None studies transfer of the selection rule across dialects.

Arabizi and dialect normalisation. Arabizi has been converted to Arabic script with rules, lexicons and sequence models [6,31]. Dialect normalisation maps dialectal text to a standard variety with rules and LLMs [7] or with dialect identification [2]; DialUp adapts models to dialect continua and dialect data to models [4], and North African dialects

have been modelled from standard languages [32]. These approaches transform all inputs.
Disagreement. Agreement between predictors has been used to estimate accuracy on unlabelled data [28,11], and querying labels where predictors disagree is the principle of query-by-committee [30].

3 Problem Formulation

Setting. Let x be a sentence with binary label $y \in \{0, 1\}$, T a deterministic normaliser from Latin-script Arabizi to Arabic script, and f_θ a frozen classifier with posterior $p_a = f_\theta(v_a)$ on view $v_0 = x$ or $v_1 = T(x)$ and decision $\hat{y}_a = \mathbf{1}[p_a > 1/2]$. A routing policy $g : \mathcal{Z} \to \{0, 1\}$ returns $\hat{y}(x) = \hat{y}_{g(z)}$, where $z = z(x)$ holds the features available at the *decision point*, after $f_\theta(x)$ and $T(x)$ and before $f_\theta(T(x))$. Let $e_a = \mathbf{1}[\hat{y}_a \neq y]$. Inference costs, relative to the mean cost of $f_\theta(x)$, are $c_0(x)$ for the first pass, $c_T(x)$ for the normaliser and $c_1(x)$ for the second pass $f_\theta(T(x))$. Because z contains p_0 and normaliser metadata, $c_0 + c_T$ is *sunk* when g is decided; only c_1 is *avoidable*.

Objective and Bayes rule. For a trade-off $\lambda \geq 0$ we seek

$$\min_g \ \mathbb{E}\big[e_{g(z)} + \lambda \left(c_0 + c_T + c_1\, g(z)\right)\big]. \tag{1}$$

The sunk term does not depend on g, so conditioning on z gives the minimiser

$$g^\star(z) = \mathbf{1}\big[B(z) > \lambda\, \bar{c}(z)\big], \qquad B(z) = \mathbb{E}[e_0 - e_1 \mid z], \quad \bar{c}(z) = \mathbb{E}[c_1 \mid z], \tag{2}$$

where B is the expected *benefit* of normalising and $\bar{c}$ the expected *avoidable* cost (proof in Appendix A). A router that reads p_0 and m costs at least $c_0 + c_T$, whereas never normalising costs c_0 and always normalising $c_T + c_1$: such routing saves compute only relative to running both views.

Disagreement factorisation (elementary). Let $D = \mathbf{1}[\hat{y}_0 \neq \hat{y}_1]$ and $W = \mathbf{1}[\hat{y}_1 = y]$. Because $e_0 = e_1$ whenever $D = 0$, and exactly one view is correct whenever $D = 1$,

$$B(z) = \pi(z)\,\big(2\rho(z) - 1\big), \qquad \pi(z) = P(D{=}1 \mid z), \quad \rho(z) = P(W{=}1 \mid D{=}1, z). \tag{3}$$

(With K classes, $B = \pi(\rho_1 - \rho_0)$.) The identity is elementary; what matters is that its factors have different data requirements: π depends only on the two views' predictions and can be estimated on *unlabelled* text, whereas ρ needs labels, but only on disagreements. Equations (2) and (3) split selective normalisation into *disagreement detection* (π), *conditional advantage estimation* (ρ) and a *cost-aware decision* ($\lambda\bar{c}$). Both estimators minimise the log-loss on the policy split ($\hat{\rho}$ on disagreements only); the plug-in rule is $\hat{g} = \mathbf{1}[\hat{\pi}(2\hat{\rho} - 1) > \lambda\bar{c}]$. Two consequences follow. (i) At $\lambda = 0$ and $\hat{\pi} > 0$, $\hat{g} = \mathbf{1}[\hat{\rho} > 1/2]$: re-estimating π can change decisions only at $\lambda > 0$. (ii) Errors of each factor are weighted by the other:

Proposition 1 (Plug-in regret). *Let $R(g)$ be the risk in Eq. (1) and $\hat{g} = \mathbf{1}[\hat{\pi}(2\hat{\rho} - 1) > \lambda\bar{c}]$, with $\bar{c}(z)$ known. Then $R(\hat{g}) - R(g^\star) \leq \mathbb{E}\big[\, |2\rho - 1|\, |\hat{\pi} - \pi| + 2\hat{\pi}\, |\hat{\rho} - \rho| \,\big]$.*

The bound (Appendix A) holds for this plug-in router with the theoretical threshold $\lambda\bar{c}$ and the true conditional cost $\bar{c}(z)$; it does not cover the variants with a threshold t_λ tuned on the policy development split, including the primary system, which are evaluated empirically. An error in $\hat{\pi}$ thus matters where the views have a clear winner, an error in $\hat{\rho}$ where they are likely to disagree, which motivates measuring them separately. Equation (3) is exact for binary labels and the 0-1 loss, and $g^\star$ minimises the population accuracy–cost objective of Eq. (1) under these assumptions. Macro-F1, our primary metric, is not a sum of per-input losses: it is evaluated empirically, and no macro-F1 optimality is claimed.

Adaptation to a new dialect. On a target distribution Q, π_Q can be re-estimated without labels; transfer of $\hat{\rho}$ assumes $P_Q(W \mid D{=}1, z) = P(W \mid D{=}1, z)$, which we test rather than assume. When ρ shifts, we model $\rho_Q(z) = \sigma\big(\operatorname{logit}\rho(z) + b\big)$ and estimate b from k labelled target disagreements.

Proposition 2 (Label efficiency of disagreement-targeted labelling). *Under this model, with π_Q known, the Fisher information about b of k target labels drawn at random is $k\,P_Q(D{=}1)\,\mathbb{E}_Q[\rho_Q(1-\rho_Q) \mid D{=}1]$, and of k labels drawn among disagreements $k\,\mathbb{E}_Q[\rho_Q(1-\rho_Q) \mid D{=}1]$; targeted labelling is $1/P_Q(D{=}1)$ times more informative.*

It concerns estimating b, not the accuracy of the resulting router.

Information access. The policy reads p_1 only on the policy split and on unlabelled target text disjoint from the evaluated fold, never at decision time; the oracle $g^{\text{or}} = \mathbf{1}[e_0{=}1 \wedge e_1{=}0]$ reads labels.

4 SelectNorm

Module A: conservative normaliser. T changes only Latin-script tokens (Algorithm 1). A token is looked up in a lexicon of Arabizi→CODA pairs built from 80% of TArC [14]; unseen tokens fall back to Tunisian character rules (*3*→ع, *7*→ح, *9*→ق, *5*→خ, *ch*→ش, word-initial vowels to alef, medial short vowels dropped). Three safeguards define T_{cons}: (i) *protection* of French and English tokens, decided from word frequencies against TArC evidence, with short or function-like tokens (*la*, *ma*, *el*) read in the language of their context; (ii) *negation*: frequent negators (*mouch*, *mech*) are mapped by a fixed list and the circumfix *ma. . . ch* is split as ما + stem + ش; (iii) *non-lexical cues*: laughter (*hhhh*, *mdr*) becomes هههه. T_{rule} (rules only) and T_{lex} (no safeguards) are controls. Each call also returns token-level metadata m (counts per branch, ambiguous characters).

Module B: benefit estimation. The decision-point features z form three groups: *conf* (maximum probability, margin and entropy of the temperature-scaled p_0), *script* (length, Latin share, digits used as letters, emoji and upper-case rates) and *trel* (the metadata m as rates per Latin token, and the share of changed tokens). $\hat{\pi}$ and $\hat{\rho}$ are logistic regressions on standardised z (gradient boosting as ablation); $\hat{\rho}$ is fitted on the disagreements of the policy split only (Algorithm 2).

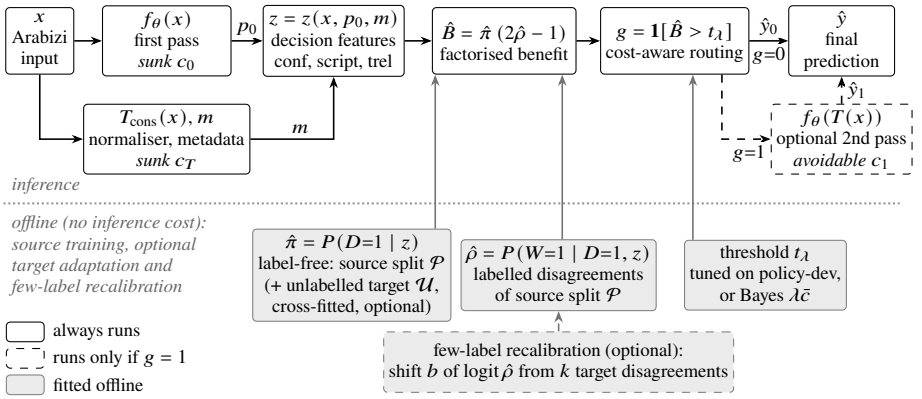


Fig. 1. SelectNorm. Upper lane (inference): $f_\theta(x)$ and $T_{\text{cons}}(x)$ always run, so c_0 and c_T are incurred before the decision g; the dashed second pass $f_\theta(T(x))$ is the only avoidable cost and runs only when $g = 1$. Lower lane (grey, offline): $\hat{\pi}$ (label-free, optionally adapted on unlabelled target text), $\hat{\rho}$ (source disagreements), the threshold t_λ and the optional few-label shift of $\hat{\rho}$.

Algorithm 1 Conservative normalisation T_{cons}

Require: sentence x, lexicon $\mathcal{L}$, rules R, frequency lists F
Ensure: $T(x)$, metadata m
$\tau \leftarrow$ tokens of x; $\kappa \leftarrow$ share of foreign tokens in τ by F vs. $\mathcal{L}$
for each Latin-script token $t \in \tau$ **do**
  if t is laughter **then** emit هههه
  else if t is foreign (if short or function-like: iff $\kappa > 1/2$) **then** keep t
  else if t is a listed negator **then** emit its fixed form
  else if $t \in \mathcal{L}$ **then** emit $\mathcal{L}(t)$
  else if t matches *ma. . . ch* **then** emit ما R(stem) ش
  else emit $R(t)$ and count ambiguous characters
  update m with the branch taken
return joined output, m

Module C: cost-aware routing. At inference (Algorithm 3) $\hat{B}(z) = \hat{\pi}(z)(2\hat{\rho}(z) - 1)$ is compared with a threshold t_λ: either the Bayes threshold $\lambda\bar{c}$ of Eq. (2), or, as for every baseline, a threshold tuned for each λ on the policy development split. No threshold is tuned on target data. The expected inference cost is $C_f + C_T + O(|z|) + \mathbb{E}[g]\, C_f$, with encoder cost C_f; only the last term is avoidable.

5 Experimental Setup

Data and classifiers. f is trained on Arabic-script TSAC [22]; the human Latin-script TSAC training comments, never seen by f, form the policy pool (Table 1). Test sets cover the same corpus (S_3), another Tunisian corpus and platform (S_4, TUNIZI [10]), Algerian (S_5, NArabizi [29,33]) and Moroccan Arabizi (S_6, MYC [17]). Mixed-script

Algorithm 2 Training

Require: f, T, labelled policy split $\mathcal{P}$; optional: unlabelled target fold $\mathcal{U}$; k labelled target disagreements $\mathcal{K}$
Ensure: $\hat{\pi}$, $\hat{\rho}$ (and shift $\hat{b}$)
for $x \in \mathcal{P} \cup \mathcal{U}$ **do** compute p_0, p_1, z, D
$\hat{\rho} \leftarrow$ fit $P(W{=}1 \mid z)$ on $\{i \in \mathcal{P} : D_i{=}1\}$
$\hat{\pi} \leftarrow$ fit $P(D{=}1 \mid z)$ on $\mathcal{P}$
  (or on $\mathcal{P} \cup \mathcal{U}$, equal total weight)
if $\mathcal{K} \neq \emptyset$ **then** ▷ few-label shift
  $\hat{b} \leftarrow \arg\max_b \ell(b)$, with $r_i = \operatorname{logit} \hat{\rho}(z_i)$:
    $\ell(b) = \sum_{\mathcal{K}} \log \mathrm{Bern}(W_i; \sigma(r_i{+}b)) - b^2/2$

Algorithm 3 Inference

Require: x, f, T, $\hat{\pi}$, $\hat{\rho}$, threshold t_λ ($\lambda\bar{c}$ or tuned on policy-dev)
Ensure: label $\hat{y}$, action g, score $\hat{B}$
$p_0 \leftarrow f(x)$; $(x', m) \leftarrow T(x)$ ▷ sunk
$z \leftarrow z(x, p_0, m)$
$\hat{B} \leftarrow \hat{\pi}(z)(2\hat{\rho}(z) - 1)$
$g \leftarrow \mathbf{1}[\hat{B} > t_\lambda]$
if $g = 1$ **then** $\hat{y} \leftarrow \mathbf{1}[f(x') > 1/2]$
else $\hat{y} \leftarrow \mathbf{1}[p_0 > 1/2]$
return $\hat{y}, g, \hat{B}$

Table 1. Data after de-duplication and near-copy removal (character 3-gram cosine ≥ 0.9 against every training or policy sentence). S_5^{AR}: human Arabic-script transliteration of S_5 [33].

Set	Variety, script	N	Pos.%	Role
TSAC-AR	Tunisian, Arabic	5,564	45.0	train f (90/10)
TSAC-LAT train	Tunisian, Latin	6,204	57.3	policy pool
S_3 TSAC-LAT test	Tunisian, Latin	1,425	55.2	test
S_4 TUNIZI	Tunisian, Latin	2,875	48.8	test
S_5 NArabizi	Algerian, Latin	729	52.0	test
S_5^{AR} NArabizi gold	Algerian, Arabic	729	52.0	RQ4
S_6 MYC-LAT	Moroccan, Latin	4,191	66.4	test

sentences (minority alphabet > 20%) are excluded. f_θ is a logistic regression on mean-pooled frozen representations of XLM-R base [5], mE5 base [34], LaBSE [9] or MARBERTv2 [1] at pinned revisions, with regularisation and a temperature [15] fitted on the 10% Arabic-script development split. No Latin-script sentence trains f. In an exploratory regime (+SYN), logged as a deviation before it was run, f also sees a synthetic Latin rendering of each training sentence.

Systems. B0 keeps x; B1 always normalises with T_{cons}; B2 routes at random at SelectNorm's selection rate (20 draws); B3 thresholds the confidence of p_0; B4 estimates $B(z) = P(\mathrm{help} \mid z) - P(\mathrm{harm} \mid z)$ directly, a post-hoc deferral rule [18]; B4b predicts each view's error [20]; B6 is the oracle; R averages both views' probabilities. SelectNorm-src uses source estimates; SelectNorm-adapt, the primary system, re-estimates $\hat{\pi}$ on unlabelled target text with two-fold cross-fitting by source sentence. All thresholds are tuned per λ on the policy development split (variant: Bayes threshold $\lambda\bar{c}$), never on a test set.

Few-label setting. Each test set is split into two folds by source sentence; to predict one fold, $k \in \{8, \ldots, 128\}$ labels are drawn from the other, at random or among its

disagreements, and re-estimate b (SelectNorm), re-fit B4, re-tune B3, or choose B0 or B1 for the fold (set-level choice). A replay of all 8,000 draws found no sentence whose own label informed its prediction. On S_5 at k=128 only 97–116 disagreements exist per fold, so 375 targeted draws are smaller than k. Each k uses 5 draws × 5 seeds and the $\lambda = 0$ threshold.

Mechanisms, statistics and costs. We measure (M1) the AUC of $\hat{\pi}$ for disagreements; (M2) the AUC of $\hat{\rho}$ for the winning view among disagreements, its calibration gap, and a *target-trained cross-validated reference* (same features and estimator family, 5-fold cross-validation on target labels); (M3) the gain over the best view of the rule $\hat{\rho} > 1/2$; (M4) the oracle headroom, against that of two classifiers trained on bootstrap resamples. Intervals are 95% percentile intervals from 1,000 seed-averaged bootstrap resamples of test sentences, without refitting. Macro-F1 is primary; results average five seeds (policy split and random draws). Differences are tested by paired bootstrap over source sentences [21,8] (10,000 resamples), Holm-corrected over the 20 main comparisons per encoder. A difference is *relevant* if it is at least one point and significant after correction; the protocol's wording requires only the unadjusted interval to exclude zero, which coincides in the primary regime (+SYN: both reported). Non-significance is not read as equivalence. Costs are relative to one pass of f on x: c_T is measured; c_0, c_1 are *estimated* by a per-encoder model $a + b \cdot$ tokens fitted to batch-1 CPU latencies and checked by end-to-end timing.

Status of analyses. Pre-specified: main comparisons and statistics, λ grid, embedding, boosting and T_{lex} ablations, few-label setting and RQ4. Exploratory, logged after the first mE5 results: T_{cov}, +SYN and M1–M4. Added in revision: mechanism intervals, sensitivity and equivalence readings, end-to-end timing, the normaliser audit and the input-dependent cost threshold.

6 Results

6.1 Selective normalisation against strong baselines (RQ1, RQ2)

Always normalising (B1) improves over the original input on the Tunisian and Algerian sets for every encoder, by 2.3–18.6 points (Table 2); on MYC it gains 8.8 points with XLM-R and loses 7.9 with LaBSE. Learned routing adds no consistent improvement. Against B4 no difference is significant and every 95% interval lies inside $[-0.63, +0.58]$, within the 1-point margin in all 16 cells (a post hoc equivalence reading); SelectNorm-adapt and -src differ by -0.25 to $+0.54$ (three significant). Against B1 and B3, SelectNorm-adapt is relevantly better in one cell (LaBSE, MYC: +5.5, +5.0), where normalising harms and thresholding does not stop it, and relevantly worse in one (XLM-R, MYC: -3.8), where it keeps the original view for about 40% of inputs although normalising is right; in the LaBSE MYC cell it is relevantly worse than B0, the better view there (-2.5). Elsewhere only 4 (B1) and 6 (B3) of 16 intervals lie within ± 1, and the other non-significant ones span -1.3 to $+2.4$, so small gains or losses cannot be excluded.

Table 2. Macro-F1 ($\lambda = 0$, mean of five seeds), f trained on Arabic script only. Bold: best deployable system per column; shaded: SelectNorm-adapt (primary). $^{\triangledown}$/$^{\blacktriangle}$: B0, B1, B3, B4 or SelectNorm-src significantly worse/better than SelectNorm-adapt (paired bootstrap, Holm-corrected $p < 0.05$, 20 comparisons per encoder). †Two passes; ‡uses test labels.

	mE5				**XLM-R**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	63.4$^{\triangledown}$	61.5$^{\triangledown}$	63.6$^{\triangledown}$	**54.1**	54.5$^{\triangledown}$	48.8$^{\triangledown}$	48.6$^{\triangledown}$	63.5$^{\triangledown}$
B1 always normalise	**82.0**	**67.2**	69.4	52.9	72.8	**64.5**	63.1	72.3$^{\blacktriangle}$
B2 random, matched rate	80.0	66.5	68.5	53.2	70.2	63.3	61.9	68.9
B3 confidence threshold	82.0	67.2	69.4	52.9	72.9	64.4	63.0	**72.3**$^{\blacktriangle}$
B4 direct gain	81.4	67.0	69.9	53.6	73.5	63.9	63.3	68.7
B4b per-view loss	81.6	67.1	69.9	53.7	**73.9**	63.9	**63.4**	68.2
SelectNorm-src	81.3	67.0	**70.0**	53.7	73.8	63.8	63.4$^{\blacktriangle}$	68.0$^{\triangledown}$
SelectNorm-adapt	81.3	66.9	69.8	53.7	73.8	63.8	63.1	68.5
Bayes threshold $\lambda\bar{c}$	81.4	66.9	69.6	53.5	73.9	64.0	63.4	69.0
R two-view ensemble†	78.9	66.5	69.7	54.4	67.3	57.2	54.7	69.7
B6 oracle‡	90.0	81.3	81.8	63.8	83.1	75.2	70.4	82.1

	LaBSE				**MARBERTv2**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	69.6$^{\triangledown}$	61.1$^{\triangledown}$	65.6$^{\triangledown}$	**63.4**$^{\blacktriangle}$	64.3$^{\triangledown}$	62.4$^{\triangledown}$	60.0	52.5$^{\triangledown}$
B1 always normalise	83.6	**69.1**	70.9	55.5$^{\triangledown}$	82.1	67.7	62.3	58.3
B2 random, matched rate	81.0	67.3	69.5	60.4	79.9	67.3	61.9	56.7
B3 confidence threshold	83.7	69.1	70.9	55.9$^{\triangledown}$	**82.3**	68.1	62.3	**58.3**
B4 direct gain	83.4	68.6	71.6	60.6	82.1	68.2	**63.0**	57.6
B4b per-view loss	83.3	68.9	**71.6**	60.5	82.1	68.1	62.8	57.6
SelectNorm-src	83.4	68.7	71.1	60.4$^{\triangledown}$	82.2	68.3	62.9	57.7
SelectNorm-adapt	83.4	68.8	71.5	60.9	82.1	**68.3**	62.9	57.7
Bayes threshold $\lambda\bar{c}$	**83.7**	68.8	71.3	60.8	82.2	68.1	62.5	57.9
R two-view ensemble†	81.7	68.4	72.1	60.5	82.3	69.7	62.4	58.4
B6 oracle‡	91.5	82.2	85.7	75.2	93.6	84.2	77.6	70.4

6.2 Accuracy and cost (RQ3)

B0 costs 1.00 and B1 0.97–1.04 passes (the normaliser adds about 0.4%). At $\lambda = 0$ SelectNorm-adapt costs 1.69–1.88 passes for mE5 and 1.43–1.88 across encoders, against 1.89–2.05 for the ensemble; raising λ lowers the cost (mE5: 1.14–1.45 at $\lambda = 0.2$) while macro-F1 falls towards B0 (Fig. 2). In Fig. 2 every router operating point lies between the single-view references (B0, B1) and R in estimated cost: routing saves compute relative to running both views, never relative to one view, and is never relevantly more accurate than the better single view (at most +0.6, not significant). At matched λ, SelectNorm and B4 differ by −0.7 to +1.2 points and by at most 0.09 passes (descriptive; only $\lambda = 0$ is tested). Timing the deployed pipeline (200 sentences per set; live decisions reproduce all 184,400 archived ones) agrees: B0 22–31 ms; B1 0.91–1.05, SelectNorm-adapt 1.31–1.97, R 1.91–2.07 passes. Thresholding with each input's own cost $\lambda c_1(x)$ (exploratory) changes

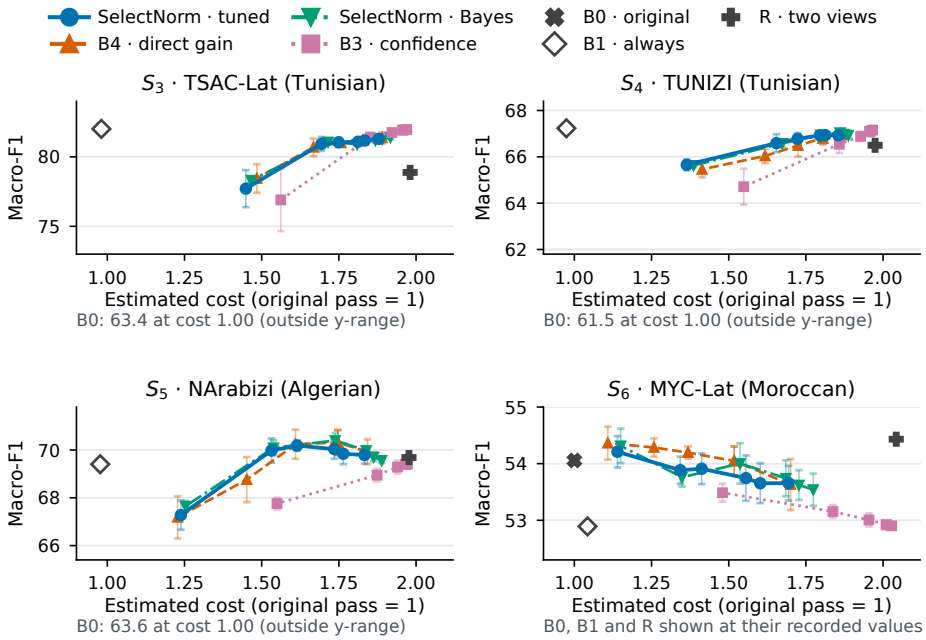


Fig. 2. Macro-F1 against *estimated* cost (mE5, CPU, batch size 1), with one encoder pass on the original input normalised to 1; measured timings are given in the text. Curves join the six evaluated settings $\lambda \in \{0, .01, .02, .05, .1, .2\}$ (no interpolation). Bars: standard deviations over seeds, not confidence intervals. Routers and B3 are charged the first pass and the normaliser (sunk) plus the second pass when selected, B1 the normaliser and one pass, R both passes. Vertical scales differ between corpora; B0 values outside the plotted range are given beneath the panels. Coinciding curves are drawn at their actual coordinates.

macro-F1 by −2.0 to +0.6 and cost by 0 to −0.11 passes, moving along the same trade-off curve (no worse on both in 29 of 80 cells, trading accuracy for cost in 48).

6.3 Which factor limits per-input routing (RQ5)

(M1) Disagreement is moderately predictable. 21–39% of inputs change label under normalisation; label-free $\hat{\pi}$ ranks them with AUC 0.62–0.74, all intervals within $[0.57, 0.77]$ (Table 3, Fig. 3). Target re-estimation helps most on MYC (mE5: 0.67 to 0.74), but the rule at $\lambda = 0$ depends on $\hat{\rho}$ alone.

(M2) Direction is weakly predictable from z with linear estimators, also with target labels. Source $\hat{\rho}$ reaches AUC 0.54–0.71 in 14 of 16 cells (Fig. 3) (upper bounds ≤ 0.76); for XLM-R the interval includes 0.5 on Algerian text and lies below it on Moroccan text ($[0.41, 0.48]$). The target-trained cross-validated reference, which uses target labels unavailable to the zero-shot router, reaches 0.53–0.68 (upper bounds ≤ 0.73; five intervals include 0.5). What changes across domains is the *level* of ρ: the win rate

Table 3. Mechanisms (T_{cons}, Arabic-script f, five seeds). M1: disagreement rate (%), AUC of target-cross-fitted $\hat{\pi}$. M2: win rate of $T(x)$ on disagreements (%), AUC of source $\hat{\rho}$ and of the target-trained cross-validated reference (unavailable to the zero-shot router), calibration gap of source $\hat{\rho}$. M3: gain of $\hat{\rho} > 1/2$ over the best view. M4: oracle headroom, minus the bootstrap-classifier reference. °: AUC interval contains 0.5; *: interval excludes 0 (95% bootstrap intervals).

		(M1) disagreement		(M2) direction				(M3) benefit		(M4) headroom	
Enc.	Set	$P(D)$	AUC_π	$P(W\|D)$	AUC_ρ	$\text{AUC}_\rho^{\text{tgt}}$	gap	Δ_{src}	Δ_{tgt}	oracle	vs. ref.
mE5	S_3	34	0.68	77	0.66	0.64	+3	-0.6*	+0.0	8.0	-0.3
	S_4	33	0.64	57	0.56	0.58	+19*	-0.3	-0.1	14.1	+4.6
	S_5	29	0.62	58	0.63	0.57°	+15*	+0.2	+0.7	12.4	+3.2
	S_6	21	0.74	47	0.54	0.56	+20*	-0.5	+0.2	9.7	+4.6
XLM-R	S_3	36	0.64	72	0.71	0.68	+3	+1.0*	+0.8	10.3	+5.9
	S_4	31	0.66	64	0.57	0.57	+11*	-0.5	+0.3	10.7	+7.6
	S_5	23	0.69	69	0.45°	0.60°	+11*	+0.3	-0.7	7.3	+2.1
	S_6	28	0.72	69	0.44	0.66	-8*	-3.3*	+1.1*	9.8	+4.5
LaBSE	S_3	30	0.69	73	0.61	0.53°	-0	+0.1	-0.1	7.9	+1.3
	S_4	34	0.65	62	0.57	0.59	+6*	-0.4	-0.4	13.0	+5.7
	S_5	34	0.65	58	0.59	0.55°	+3	+0.4	+0.2	14.9	+7.5
	S_6	31	0.68	33	0.57	0.64	+16*	-2.6*	+0.8*	11.8	+5.3
MARBERTv2	S_3	39	0.69	70	0.63	0.60	+1	+0.1	-0.3	11.5	+0.8
	S_4	37	0.71	60	0.61	0.64	+13*	+0.4	+2.8*	16.5	+2.0
	S_5	31	0.73	57	0.63	0.54°	+13*	+0.2	+1.1	15.3	+3.6
	S_6	30	0.71	59	0.55	0.63	+4*	-0.3	+1.1*	12.1	-0.7

ranges from 33% to 77%, and source $\hat{\rho}$ is miscalibrated out of domain (gaps up to +20 points; intervals exclude zero in 11 of 12 cells).

(M3, M4) Benefit and headroom. The rule $\hat{\rho} > 1/2$ gains −3.3 to +1.0 points over the best view with source estimates (one interval above zero, three below) and −0.7 to +2.8 with the target-trained reference (upper bounds ≤ +3.9): most of the gap to the oracle remains when the router is fitted on target labels. The oracle gains 7–17 points over the best view, but two classifiers differing only in a bootstrap resample of their training data show 3–15 points of such headroom; this reference depends on how the classifiers are made (2–6 with 80% subsamples, 5–22 with weaker regularisation) and is not a decomposition.

Linguistic conditions. On Tunisian and Algerian sets, normalising helps less on inputs with many French or English tokens or low lexicon coverage (except XLM-R on TUNIZI for French/English); negation and length show no consistent pattern; on MYC every condition changes sign with the encoder.

6.4 Few target labels (RQ6), ablations and classifier regime

With k labelled target sentences, re-estimating the shift of $\hat{\rho}$ changes macro-F1 relative to zero-shot SelectNorm-adapt by −1.4 to +1.5 points across encoders, regimes, sets and k (mean −0.01); although targeted labels are $1/P(D{=}1)$ times more informative about the shift (Proposition 2), the resulting macro-F1 is within −0.1 to +0.2 points of random labels at every k (averaged, descriptive). In the Arabic-script regime SelectNorm beats a set-level choice between B0 and B1 by 2.6–2.9 points on average for $k \leq 16$ and by 0.3 at $k = 128$, consistent with M2.

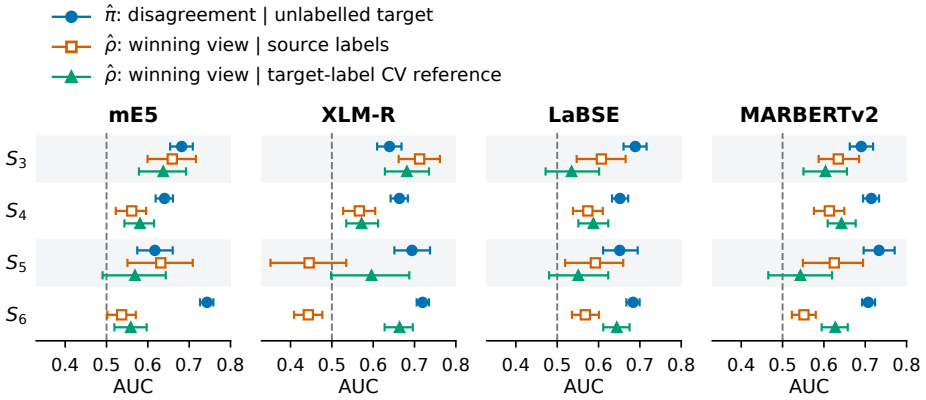


Fig. 3. AUC by encoder and test set (values of Table 3). Circles: disagreement prediction $\hat{\pi}$, fitted on unlabelled target text; squares: winning-view prediction $\hat{\rho}$ among disagreements, trained on source labels; triangles: target-label reference, same features and estimator cross-validated on target labels, which are unavailable to the zero-shot router. Bars: 95% bootstrap intervals over test sentences, estimators not refitted; dashed line: AUC = 0.5. Tasks, evaluation populations and information access differ, so the comparison is descriptive; the reference is not a theoretical ceiling.

Ablations (mE5, $\lambda = 0$). The safeguards set the quality of the views the router chooses between. Without them, always normalising loses 4.2–6.2 points (T_{lex}) and 5.2–10.6 (T_{rule}) on every set, and routing over T_{lex} stays 0.6–4.6 points below routing over T_{cons}. Keeping out-of-lexicon tokens in Latin script (T_{cov}) reduces disagreements by about a third; it is worse than T_{cons} on the Tunisian sets and slightly better on S_5 and S_6. Routing components matter less. Removing $\hat{\pi}$ changes macro-F1 by at most 0.1 point, as Sect. 3 implies for $\lambda = 0$, dropping its target re-estimation by at most 0.2, removing one feature group by −0.5 to +0.7, and gradient boosting by −0.6 to +0.7. Two label-free simplifications ($\hat{\pi}$ alone; confidence features alone) stay within 0.2 points of always normalising on every set and 0.6–0.7 below the full router on S_6.

Human transliterations (RQ4) and +SYN. On NArabizi, B1 agrees with f's prediction on the human transliteration for 73–77% of inputs (B0: 61–67%; SelectNorm-adapt: 73–76%), on which f reaches 65–77 macro-F1. In +SYN the original view improves and the win rate on disagreements nears 50%; SelectNorm-adapt is relevantly better than the better view once with Holm correction (MARBERTv2, S_4, +1.1), twice more unadjusted (mE5, S_3 +2.1, S_4 +1.4), and relevantly worse in two MYC cells under both (LaBSE −2.4, MARBERTv2 −1.8); it is never significantly better than B4 (worse once, −0.4).

6.5 Qualitative examples

In Table 4 (hand-picked), TSAC 796 shows the negator list and the protection of *role*, TSAC 86 the rule fallback. In both NArabizi sentences *le* becomes ⵏ (*no*); the router selects $T(x)$, right in sentence 2 and wrong in 4609.

Table 4. Test sentences with archived predictions (mE5, $\lambda = 0$, SelectNorm-adapt): x above $T_{\text{cons}}(x)$; gold y (1 = positive); posteriors p_0, p_1; routing g and prediction $\hat{y}$, seeds 0–4. Sources: TSAC test line (LGPL-3.0); NArabizi `sent_id` (UD, CC BY-SA 4.0); MYC row, text withheld (no licence).

Source, effect	Original x / normalised $T_{\text{cons}}(x)$	y	p_0	p_1	seeds 0–4
TSAC neg. 796	*mech 5arej 3lik el role*	0	.565	.127	g 11111
help	role مش خارج عليك ال				$\hat{y}$ **00000**
TSAC pos. 86	*Metrobi mouch kif ijale4 la5er*	1	.806	.004	g 11111
harm	مترﺑﻲ موش كيف يجلذ لخر				$\hat{y}$ **00000**
NArabizi 2	*salem 3alikoum inchalah le pondium et les midailes d'or*	1	.175	.842	g 11111
help	سلام عليكوم ان شاء الله لا بنديوم ات لس ميديلس د ’ اور				$\hat{y}$ 11111
NArabizi 4609	*oui le Maroc aktar jahiziya li had tasfiyat la 3ala lmostawa tandimi*	1	.519	.088	g 11111
harm	وي لا Maroc اكتر جهيزيا اللي حد تسفييت لا علي لمستوا تنديمي				$\hat{y}$ **00000**
MYC 9079	[withheld] two *ma. . . ch* negations split as ما . . . ش,	1	.645	.250	g **00**111
harm (s2–4)	letter elongations squeezed, emoji kept				$\hat{y}$ **11000**
MYC 14543	[withheld] two Latin-script loanwords protected,	1	.480	.302	g **00001**
both wrong	negated verb split, emoji kept				$\hat{y}$ **00000**

7 Discussion

The factorisation separates disagreement detection from conditional advantage estimation, making their contributions to routing performance measurable. Transfer limits routing only for XLM-R on S_5 and S_6, where target estimates turn a loss of up to 3.3 points into a small gain; elsewhere the limitation persists with target labels and with 16 embedding components (-1.2 to $+1.1$). The win rate on disagreements varies more between encoder–dialect pairs (33–77%) than the features resolve within a set; with 64–128 target labels, a set-level choice is on average within 0.3–0.7 points of SelectNorm, without a guarantee for every configuration. These are limits of the features and estimators evaluated, not evidence of intrinsic unpredictability. The results support corpus- and encoder-specific view selection with a safeguarded normaliser in this frozen-encoder setting; dialect, corpus, platform, content and normaliser effects are not isolated.

8 Limitations

Task and models. All experiments use binary sentiment (the K-class form needs two win rates). Encoders are frozen and heads linear, and the conclusions are restricted to this setting; fine-tuning (not done; +SYN is a partial probe) could change how often normalisation helps and how predictable the winning view is. *Normaliser.* T uses Tunisian resources; its safeguards were tuned on held-out TArC, so its intrinsic accuracy is *not independently validated*; it is lower on NArabizi and unmeasured on MYC. *Data.* Cross-corpus differences mix dialect with platform, topic, period and label distribution; we

do not attribute them to dialect alone. TUNIZI, the NArabizi sentiment layer, MYC and MARBERTv2 state no licence; they are used for research only and not redistributed. *Development.* A character n-gram pilot whose test results we inspected motivated the few-label setting. *Analysis.* Mechanism intervals do not cover estimator refitting; the target-trained reference is not an upper bound on achievable performance. Features computed after $f(T(x))$ were not studied. Equivalence readings are post hoc and uncorrected for multiplicity; the examples are hand-picked. *Costs.* Estimated from a token-length model and checked by timing on one 4-core CPU; relative costs on accelerators differ.

9 Conclusion

We framed per-input normalisation of Arabizi for frozen encoders as a cost-aware choice between two views and factorised its expected benefit into disagreement and conditional advantage. Normalisation improves macro-F1 on Tunisian and Algerian Arabizi; on Moroccan text its effect depends on the encoder. SelectNorm performs on par with a direct-gain router but does not consistently exceed the better single view; because the first pass is sunk, routing saves compute only relative to running both views. Disagreement is moderately predictable from decision-point features, whereas the conditional advantage is weakly predictable with the features and linear estimators evaluated, also with target labels. The normaliser's safeguards matter more than the routing components. The results support corpus- and encoder-specific view selection in this frozen-encoder setting.

Code and Reproducibility. Code, the pre-registered protocol, logs, per-example predictions, figure data and pinned revisions are at [GITHUB_REPOSITORY_URL]; corpora are not redistributed. A script recomputes every test of Table 2. Quoted sentences come from the Universal Dependencies release of NArabizi [29] (CC BY-SA 4.0) and from TSAC [22] (LGPL-3.0).

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References

1. Abdul-Mageed, M., Elmadany, A., Nagoudi, E.M.B.: ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In: Proceedings of ACL-IJCNLP. pp. 7088–7105 (2021)
2. Alhafni, B., Al-Towaity, S., Fawzy, Z., Nassar, F., Eryani, F., Bouamor, H., Habash, N.: Exploiting dialect identification in automatic dialectal text normalization. In: Proceedings of the Second Arabic Natural Language Processing Conference. pp. 42–54 (2024)
3. Artetxe, M., Goswami, V., Bhosale, S., Fan, A., Zettlemoyer, L.: Revisiting machine translation for cross-lingual classification. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. pp. 6489–6499 (2023)
4. Bafna, N., Chang, E., Robinson, N.R., Mortensen, D.R., Murray, K., Yarowsky, D., Sirin, H.: DialUp! modeling the language continuum by adapting models to dialects and dialects to models. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 20188–20233 (2025)

5. Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., Stoyanov, V.: Unsupervised cross-lingual representation learning at scale. In: Proceedings of ACL. pp. 8440–8451 (2020)
6. Darwish, K.: Arabizi detection and conversion to Arabic. In: Proceedings of the EMNLP 2014 Workshop on Arabic Natural Language Processing (ANLP). pp. 217–224 (2014)
7. Dimakis, A., Pavlopoulos, J., Anastasopoulos, A.: Dialect normalization using large language models and morphological rules. In: Findings of the Association for Computational Linguistics: ACL 2025. pp. 23696–23714 (2025)
8. Dror, R., Baumer, G., Shlomov, S., Reichart, R.: The hitchhiker's guide to testing statistical significance in natural language processing. In: Proceedings of ACL. pp. 1383–1392 (2018)
9. Feng, F., Yang, Y., Cer, D., Arivazhagan, N., Wang, W.: Language-agnostic BERT sentence embedding. In: Proceedings of ACL. pp. 878–891 (2022)
10. Fourati, C., Haddad, H., Messaoudi, A., BenHajhmida, M., Ben Elhaj Mabrouk, A., Naski, M.: Introducing a large Tunisian Arabizi dialectal dataset for sentiment analysis. In: Proceedings of the Sixth Arabic Natural Language Processing Workshop. pp. 226–230 (2021)
11. Garg, S., Balakrishnan, S., Lipton, Z.C., Neyshabur, B., Sedghi, H.: Leveraging unlabeled data to predict out-of-distribution performance. In: International Conference on Learning Representations (2022)
12. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems. pp. 4878–4887 (2017)
13. Geifman, Y., El-Yaniv, R.: SelectiveNet: A deep neural network with an integrated reject option. In: Proceedings of the 36th International Conference on Machine Learning. PMLR, vol. 97, pp. 2151–2159 (2019)
14. Gugliotta, E., Dinarelli, M.: TArC: Tunisian Arabish corpus, first complete release. In: Proceedings of the Thirteenth Language Resources and Evaluation Conference. pp. 1125–1136 (2022)
15. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proceedings of ICML. pp. 1321–1330 (2017)
16. Gupta, N., Narasimhan, H., Rawat, A.S., Jitkrittum, W., Menon, A.K., Kumar, S.: Language model cascades: Token-level uncertainty and beyond. In: International Conference on Learning Representations (2024)
17. Jbel, M., Jabrane, M., Hafidi, I., Metrane, A.: Sentiment analysis dataset in Moroccan dialect: Bridging the gap between Arabic and Latin scripted dialect. Language Resources and Evaluation **59**(2), 1401–1430 (2025). https://doi.org/10.1007/s10579-024-09764-6, published online 2024; preprint arXiv:2303.15987
18. Jitkrittum, W., Gupta, N., Menon, A.K., Narasimhan, H., Rawat, A.S., Kumar, S.: When does confidence-based cascade deferral suffice? In: Advances in Neural Information Processing Systems (2023)
19. Kang, D., Lee, G.G., Kim, H.: Learning when to translate for multilingual reasoning. arXiv:2606.02465 (2026)
20. Kim, I., Kim, Y., Kim, S.: Learning loss for test-time augmentation. In: Advances in Neural Information Processing Systems. vol. 33, pp. 4163–4174 (2020)
21. Koehn, P.: Statistical significance tests for machine translation evaluation. In: Proceedings of EMNLP. pp. 388–395 (2004)
22. Medhaffar, S., Bougares, F., Estève, Y., Hadrich-Belguith, L.: Sentiment analysis of Tunisian dialects: Linguistic resources and experiments. In: Proceedings of the Third Arabic Natural Language Processing Workshop. pp. 55–61 (2017)
23. Moosa, I.M., Akhter, M.E., Habib, A.B.: Does transliteration help multilingual language modeling? In: Findings of the Association for Computational Linguistics: EACL 2023. pp. 670–685 (2023)

24. Mozannar, H., Sontag, D.: Consistent estimators for learning to defer to an expert. In: Proceedings of the 37th International Conference on Machine Learning. PMLR, vol. 119, pp. 7076–7087 (2020)
25. Muller, B., Anastasopoulos, A., Sagot, B., Seddah, D.: When being unseen from mBERT is just the beginning: Handling new languages with multilingual language models. In: Proceedings of NAACL-HLT. pp. 448–462 (2021)
26. Noorani, S., Kiyani, S., Pappas, G., Hassani, H.: Disagreement-guided weak-to-strong routing. arXiv:2610.22245 (2026)
27. Ong, I., Almahairi, A., Wu, V., Chiang, W.L., Wu, T., Gonzalez, J.E., Kadous, M.W., Stoica, I.: RouteLLM: Learning to route LLMs from preference data. In: International Conference on Learning Representations (2025)
28. Platanios, E.A., Blum, A., Mitchell, T.M.: Estimating accuracy from unlabeled data. In: Proceedings of the Thirtieth Conference on Uncertainty in Artificial Intelligence. pp. 682–691 (2014)
29. Seddah, D., Essaidi, F., Fethi, A., Futeral, M., Muller, B., Ortiz Suárez, P.J., Sagot, B., Srivastava, A.: Building a user-generated content North-African Arabizi treebank: Tackling hell. In: Proceedings of ACL. pp. 1139–1150 (2020)
30. Seung, H.S., Opper, M., Sompolinsky, H.: Query by committee. In: Proceedings of the Fifth Annual Workshop on Computational Learning Theory (1992). https://doi.org/10.1145/130385.130417
31. Shazal, A., Usman, A., Habash, N.: A unified model for Arabizi detection and transliteration using sequence-to-sequence models. In: Proceedings of the Fifth Arabic Natural Language Processing Workshop. pp. 167–177 (2020)
32. Toughrai, Y., Smaïli, K., Langlois, D.: Modeling North African dialects from standard languages. In: Proceedings of The Third Arabic Natural Language Processing Conference. pp. 375–383 (2025)
33. Touileb, S., Barnes, J.: The interplay between language similarity and script on a novel multi-layer Algerian dialect corpus. In: Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021. pp. 3700–3712 (2021)
34. Wang, L., Yang, N., Huang, X., Yang, L., Majumder, R., Wei, F.: Multilingual E5 text embeddings: A technical report. arXiv preprint arXiv:2402.05672 (2024)

A Proofs

Bayes rule and factorisation. Given z, the conditional risk of $g(z) = a$ is $\mathbb{E}[e_a \mid z] + \lambda\,\mathbb{E}[c_0 + c_T \mid z] + \lambda\bar{c}(z)\,a$, so $a = 1$ is strictly better iff $B(z) > \lambda\bar{c}(z)$; pointwise minimisation minimises Eq. (1). If $D = 0$, $e_0 = e_1$; if $D = 1$ and $y \in \{0, 1\}$, $e_0 = W$, $e_1 = 1 - W$; hence $e_0 - e_1 = D(2W - 1)$, which gives Eq. (3). For K classes, $e_0 - e_1 = D(\mathbf{1}[\hat{y}_1{=}y] - \mathbf{1}[\hat{y}_0{=}y])$. □

Proposition 1. Where $\hat{g} \neq g^\star$, $\hat{B} - \lambda\bar{c}$ and $B - \lambda\bar{c}$ differ in sign (or $\hat{B} = \lambda\bar{c}$), so the conditional excess risk is $|B - \lambda\bar{c}| \leq |\hat{B} - B|$. Hence $R(\hat{g}) - R(g^\star) \leq \mathbb{E}|\hat{B} - B|$, and $\hat{B} - B = (\hat{\pi} - \pi)(2\rho - 1) + 2\hat{\pi}(\hat{\rho} - \rho)$ with the triangle inequality concludes. □

Proposition 2. With $q_i = \sigma(\mathrm{logit}\,\hat{\rho}(z_i) + b)$, a labelled target example has log-likelihood $D_i[W_i \log q_i + (1 - W_i)\log(1 - q_i)]$ and Fisher information $D_i\,q_i(1 - q_i)$ about b, with expectation $P_Q(D{=}1)\,\mathbb{E}_Q[q(1 - q) \mid D{=}1]$ under random sampling and $\mathbb{E}_Q[q(1 - q) \mid D{=}1]$ among disagreements, which are observable without labels; information adds over draws. □